

Sparse Vector Reconstruction from Distance Spectrum Using Soft Information

Magali Salom^{1,2}, Nicolas Sendrier², Valentin Vasseur¹

¹ Thales, Gennevilliers, France

² Inria, Paris, France

{magali.salom,nicolas.sendrier}@inria.fr, valentin.vasseur@thalesgroup.com

Abstract. QC-MDPC based schemes feature secret sparse binary vectors in a cyclic polynomial ring. When those vectors are sparse enough, they can be reconstructed from their distance spectrum, that is the set of all distances between the coordinates of the non-zero coefficients. In this work, we revisit the reconstruction algorithms and we explore to what extent a secret sparse vector can be recovered from a partial knowledge of its distance spectrum. In particular, we show how to efficiently use reliability (soft information) in the reconstruction process. Another aspect of this work is to investigate which kind of side-channel leaks information about the distance spectrum and to understand the models that enable us to quantify the reliability on leaking data depending on the amount of side-channel observations (or queries). For instance, we show that for BIKE level 1, assuming that a side-channel leaks information about the syndrome weight, using soft information in the reconstruction process reduces the number of queries by a factor 10. Our technique can also be applied to HQC, which also features sparse secret vector, with similar figures, assuming there exists a side-channel leaking relevant information, the error weight in the case of HQC.

Keywords: code-based cryptography, QC-MDPC codes, BIKE, HQC, key recovery attack, distance spectrum, side-channel attack

1 Introduction

Some recent code-based cryptosystems (e.g. BIKE or HQC) involve secret sparse vectors in a cyclic polynomial ring. In such situations, the knowledge of the distance spectrum—the set of all distances between the coordinates of the non-zero coefficients [4]—allows the efficient reconstruction of the secret. Information on the distance spectrum may leak from a side-channel, e.g. reaction attack [4] or timing attack [2], which therefore allows key recovery.

The general scenario is a known plaintext side-channel attack, the attacker can make multiple queries to a decryption device which is assumed to leak information about the spectrum of a given fixed sparse secret. With sufficiently many queries the attacker can recover the full spectrum and efficiently reconstruct the secret. With less queries, the attacker only has a partial knowledge of the spectrum and the reconstruction, though harder, may still remain feasible.

In this paper, we revisit the reconstruction algorithm, in particular we propose and implement a variant able to handle a soft input spectrum, allowing us to efficiently deal with reliability information. Together with an accurate leakage model we are able to quantify the gain provided by such an algorithm. In the case of BIKE-1, the number of queries is reduced by a factor 10. Our study is made in an idealized abstract model in which the adversary has access to the Hamming weight of the syndrome used in BIKE’s decapsulation procedure. In a more realistic scenario, the information leaking from the side-channel may only have a small correlation with the syndrome weight, possibly with extra noise. This will increase the number of queries to get a full or partial spectrum, but we believe that the gain factor between hard and soft input reconstructions will be of the same order.

1.1 Overview

The distance spectrum $\text{Sp}(g)$ of a sparse vector $g \in \mathcal{R} = \mathbb{F}_2[x]/(x^r - 1)$ is the set of all distances between its 1’s. For QC-MDPC-based encryption schemes, a sparse error pattern $e = (e_0, e_1) \in \mathcal{R}^2$ is encrypted using a public key $\mathbf{pk}$ into $c = \text{Enc}(e, \mathbf{pk})$ and later decrypted using a sparse private key $\mathbf{sk} = (h_0, h_1) \in \mathcal{R}^2$ as $e = \text{Dec}(c, \mathbf{sk})$. An adversary may query $\text{Dec}(\text{Enc}(e, \mathbf{pk}), \mathbf{sk})$ with known e and observe a numerical value³ $f(e_0, h_0)$ such that for all δ that may appear in a distance spectrum, i.e. $1 \leq \delta \leq r/2$, the (conditional) expectation $\mathbb{E}(f(e_0, h_0) \mid \delta \in \text{Sp}(e_0))$ is different depending on whether δ belongs to $\text{Sp}(h_0)$ or not. Concretely, for a given public/private key pair $\mathbf{pk}/\mathbf{sk}$:

- (1) sample a set $\mathcal{E} \subset \mathcal{R}^2$ of error patterns ;
- (2) for all $e \in \mathcal{E}$, query the decryption oracle on $\text{Enc}(e, \mathbf{pk})$ and observe $f(e_0, h_0)$;
- (3) for all $1 \leq \delta \leq r/2$, let $\mathcal{E}_\delta = \{e \in \mathcal{E} \mid \delta \in \text{Sp}(e_0)\}$ and decide whether $\delta \in \text{Sp}(h_0)$ depending on the value of

$$F_\delta(\mathcal{E}, (h_0, h_1)) = \frac{1}{|\mathcal{E}_\delta|} \sum_{e \in \mathcal{E}_\delta} f(e_0, h_0). \quad (1)$$

Hopefully, if the sample set $\mathcal{E}$ is large enough, the adversary may recover the full spectrum of h_0 and then reconstruct h_0 . This framework was first used in [4] with $f(e_0, h_0) = 1$ in case of decoding failure and 0 otherwise. Later, it was used in [2], either with $f(e_0, h_0) = i$ the number of iterations to succeed in the bit-flipping decoder, or with $f(e_0, h_0) = |e_0 h_0 + e_1 h_1|$ the syndrome weight.

For instance, we plotted in Figure 1 the function $\delta \mapsto F_\delta$ (top) and the density function of the distribution of F_δ (bottom) when the observable is the syndrome weight. Other observables, e.g. failure or timing, give similar diagrams but require a substantially higher number of decryption queries. Dots in Figure 1 are red if $\delta \notin \text{Sp}(h_0)$ and blue otherwise. We remark that the values of F_δ

³ the attack may target either h_0 using e_0 , or h_1 using e_1 , without loss of generality we assume that the first block is targeted

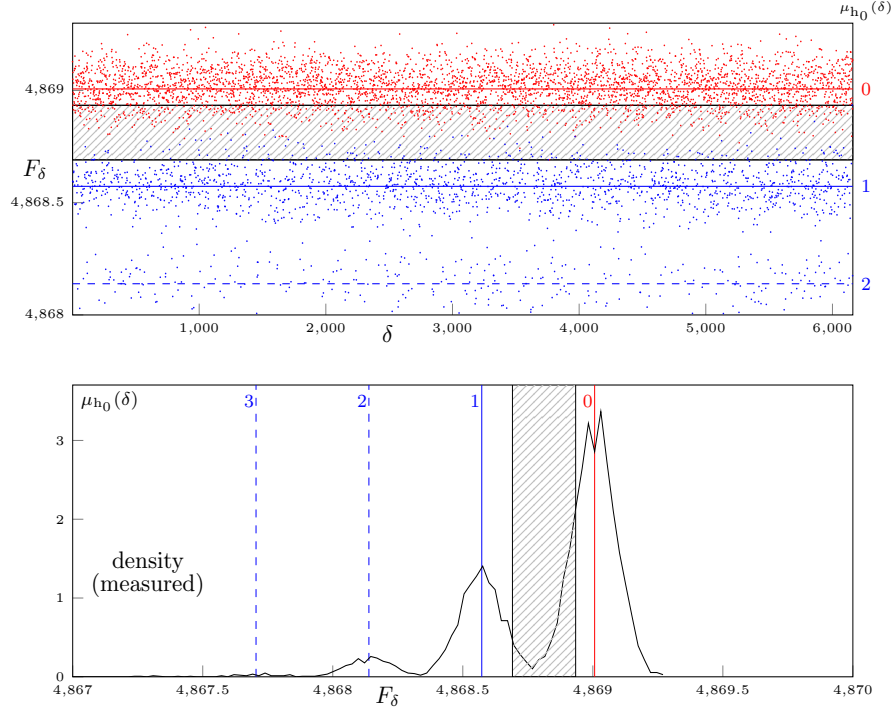


Fig. 1. $F_\delta(\mathcal{E}, (h_0, h_1))$ is the average of the observable $f(e_0, h_0) = |e_0 h_0 + e_1 h_1|$ over errors in $\mathcal{E}$ that contain distance δ (see Equation 1). Here, $|\mathcal{E}| = 10^6$ for BIKE-1 parameters (see Table 1). The red lines correspond to distances that do not appear in h_0 (multiplicity $\mu_{h_0}(\delta) = 0$), blue lines to distances appearing once ($\mu_{h_0}(\delta) = 1$), and dashed lines to distances with higher multiplicities ($\mu_{h_0}(\delta) \geq 2$). The hatched area indicates distances that are the least reliable when using a partial spectrum. **Upper plot:** F_δ for all distances δ . Colors indicate whether the distance appears in the spectrum (blue) or not (red). **Bottom plot:** Measured density of F_δ .

concentrate mostly in two stripes, red when $\delta \notin \text{Sp}(h_0)$ and blue⁴ when $\delta \in \text{Sp}(h_0)$. The adversary, which is oblivious to colors but knows by analysis the red and blue lines, must guess the color of each dot depending on where it lies in the diagram. Finally, those diagrams emphasize the impact of multiplicity—we denote $\mu_g(\delta)$ the number occurrences of the distance δ between the 1 coordinates of g —on the modeling of the density distribution function of F_δ .

Recovery from Full or Partial Spectrum: current key recovery techniques are successful when the full spectrum has been correctly guessed, but they are vulnerable to faulty data. On the other hand, reconstruction algorithms can easily be adapted to situations where the status, inside or outside the spectrum, of a

⁴ the additional dashed lines correspond to distances that appear multiple times in h_0

small proportion of the distances is unknown—or deliberately ignored to avoid mistakes. Typically, the least reliable distances δ are such that F_δ lies inside the hatched area in Figure 1. Those distances can be ignored without impairing the reconstruction as long as there are not too many of them.

Soft Information: going further involves soft information and requires some kind of modeling. For each δ , the observation F_δ is transformed into a probability distribution $(p_\nu)_{\nu \geq 0}$ where p_ν is the probability, given the observation F_δ , to have δ appearing in $\text{Sp}(h_0)$ with multiplicity ν . The reconstruction algorithm can then be adapted to properly and efficiently handle this reliability information.

1.2 Related Works

For code-based cryptography, the distance spectrum and the above framework were introduced in [4], in which the observable is the decoding failure rate. This paper also features a backtrack search reconstruction algorithm to obtain a key from its full spectrum. Variants of this attack with different observables—namely timing and syndrome weight—were proposed in [2] and variants of the key reconstruction algorithm were proposed in [9,8].

Key recovery from partial spectrum was first explored in [9]. In contrast with other works, their reconstruction technique requires both spectra, $\text{Sp}(h_0)$ and $\text{Sp}(h_1)$. The threshold to ignore distances is determined empirically and a reduction of 20% on the number of queries is claimed. Recovery from partial spectrum is also considered in [7], using the reconstruction technique of [8] which features a breadth-first search—rather than depth-first. The ignored distances, corresponding to the hatched region in Figure 1, is determined from a model of the density function. The model is a combination of Gaussian distributions whose parameters are determined empirically.

Key recovery from spectrum using soft information has been considered in [5]. The main focus of that paper is not reconstruction, but still it suggests some improvements for the backtrack search and also provides some guidelines for the use of reliability information. The latter was not implemented in [5], but the authors estimate that using soft information could reduce the number of queries by a factor 2 to 4.

Finally, let us mention that this reconstruction problem has appeared in other fields, e.g. genetics or crystallography, and is sometimes referred to as the “turnpike” or “beltway” problem [10,1,6]. The context and parameters are rather different though.

1.3 Contributions

We consider a scenario in which the attacker derives the key spectrum, possibly with reliability information, from the observation of the syndrome weight.

- *An Improved Algorithm.* We revisit the key reconstruction algorithm from a full spectrum. We speed up the backtrack search technique by moving two

steps at a time in the search tree. We also improve the pruning condition (*a.k.a.* consistency check) by introducing the concept of extension of a polynomial relatively to a spectrum. Overall, this improves on previous reconstruction techniques. With BIKE-1 parameters obtaining the full spectrum requires about $2 \cdot 10^6$ decryption queries with access to the syndrome-weight oracle.

- *Modeling the Observed Distribution.* We remark that the distribution of F_δ features a better discrimination between the key spectrum multiplicities when the observable is weighted with the multiplicity of δ in the error spectrum, for a given sample set $\mathcal{E}$ of error patterns:

$$F_\delta(\mathcal{E}, (h_0, h_1)) = \frac{1}{\sum_{e \in \mathcal{E}} \mu_{e_0}(\delta)} \sum_{e \in \mathcal{E}} \mu_{e_0}(\delta) f(e_0, h_0). \quad (2)$$

The usual model for the probability density function of F_δ is a combination of several normal distributions [5,7], one for each multiplicity, which is empirically determined, see Figure 1 (bottom). When the observable is the syndrome weight, we obtain a model for (2) with proven coefficients for the combination and proven means for the normal distributions. The standard deviations are still determined empirically but, interestingly, seem to be the same for all multiplicities.

- *Reconstruction from Partial Spectrum.* We show how our algorithm can be adapted to handle a partial spectrum. In particular the above model is used to determine the least reliable distances. To perform correctly, our algorithm must know most of the distances that do not belong to the spectrum but needs only a very small part of the spectrum. For BIKE-1 parameters, the number of queries drops to 700 000, about one third of the amount required for a full spectrum reconstruction.
- *Reconstruction from Soft Information.* We adapt our reconstruction algorithm to soft information, using a methodology close to the guidelines given in [5]. From our model for (2), we derive the multiplicity distribution for each δ and a score function which measures the chances for a node in the search tree to be in the path of a solution. This allows us to efficiently browse and prune the search space. For BIKE-1 parameters, the number of queries drops to 200 000, about one tenth of the amount required for a full spectrum reconstruction.
- *Generalization.* Though our primary target is BIKE in an ideal model where the adversary can directly observe the syndrome weight, we show that our technique can be adapted to other schemes, e.g. HQC, and to other side-channels as long as the leakage is correlated to the syndrome weight.

1.4 Artifact

C code to reproduce the simulations is available at https://github.com/msalom/key_rec_spectrum. All the experiments were performed on a laptop featuring an Intel Core Ultra 7 165U with a 4.9 GHz maximum clock frequency. The C implementation is compiled with gcc version 13.3.0 using -Ofast.

1.5 Paper Organization

We first give some background results and definitions in §2, then present all variations of our reconstruction algorithm from a full, partial and soft spectrum in §3, §4 and §5 respectively. We give in §6 our model for the probability distribution of our observable. Finally, we explain in §7 how the framework can be extended to other observables and schemes, and we give the numerical results of some of our experiments.

2 Preliminaries

NOTATION

$\mathbb{F}_2$:	the binary finite field
$\mathcal{R}$:	the cyclic polynomial ring $\mathbb{F}_2[x]/(x^r - 1)$, r an odd prime $g, h \in \mathcal{R}$
g_i :	$0 \leq i < r$, the i -th coefficient of $g = \sum_{0 \leq i < r} g_i x^i$
$ g $:	the Hamming weight of g
$\text{supp}(g)$:	$\{i \mid 0 \leq i < r, g_i = 1\}$
$i \in g$:	$i \in \text{supp}(g)$
$h \subset g$:	$\text{supp}(h) \subset \text{supp}(g)$
$f = h \cup g$:	$f \in \mathcal{R}$ such that $\text{supp}(f) = \text{supp}(h) \cup \text{supp}(g)$
$f = h \cap g$:	$f \in \mathcal{R}$ such that $\text{supp}(f) = \text{supp}(h) \cap \text{supp}(g)$

2.1 Distance Spectrum

Definition 1 (Distance Spectrum). Let $\mathcal{R} = \mathbb{F}_2[x]/(x^r - 1)$ with r an odd prime integer, let $h \in \mathcal{R}$, and let δ be an integer such that $1 \leq \delta \leq r/2$.

- The coordinate distance is defined for all $0 \leq i, j < r$ as

$$d(i, j) = \min(i - j \bmod r, j - i \bmod r), 1 \leq d(i, j) < r/2.$$

- The multiplicity of δ in h is defined as

$$\mu_h(\delta) = |\{(i, j) \mid h_i = h_j = 1, 0 \leq i < j < r, d(i, j) = \delta\}|.$$

- The distance spectrum of $h \in \mathcal{R}$ is defined as

$$\text{Sp}(h) = (\mu_h(\delta))_{1 \leq \delta < r/2} \in \mathbb{N}^{\lfloor r/2 \rfloor}.$$

Originally [4] the distance spectrum is defined as a subset of $\{1, \dots, (r-1)/2\}$. We use multiplicity vectors instead and we extend the set notation as follows: for any $S = (S_\delta)_{1 \leq \delta < r/2}$ and $S' = (S'_\delta)_{1 \leq \delta < r/2}$ in $\mathbb{N}^{\lfloor r/2 \rfloor}$, we denote $\delta \in S$ if $S_\delta > 0$ and $S \subset S'$ if $S_\delta \leq S'_\delta$ for all δ .

Proposition 1 (Multiplicity Distribution [11]). For all integers δ , $1 \leq \delta < r/2$, and $\nu \geq 0$, we have

$$\rho_\nu(r, d) = \Pr[\mu_h(\delta) = \nu] = \frac{\binom{d}{\nu} \binom{r-d-1}{d-\nu-1}}{\binom{r-1}{d-1}}$$

for $h \in \mathcal{R}$ uniformly distributed of Hamming weight d .

Some Basic Properties.

- Monotony (strict): $h \subsetneq g \Rightarrow \text{Sp}(h) \subsetneq \text{Sp}(g)$, $g, h \in \mathcal{R}$.
(strict because the coefficients of $\text{Sp}(h)$ add to $|h| \cdot (|h| - 1)/2$)
- Invariance under rotation: $\text{Sp}(x^i g) = \text{Sp}(g)$, $g \in \mathcal{R}$ and $0 \leq i < r$.
- Invariance by mirroring: $\text{Sp}(\bar{g}) = \text{Sp}(g)$, $g \in \mathcal{R}$ and $\bar{g} = \sum_{0 \leq i < r} g_{r-1-i} x^i$.

2.2 QC-MDPC Codes – BIKE

The key encapsulation mechanism BIKE derives from a Niederreiter-like public key encryption scheme based on QC-MDPC (Quasi-Cyclic Moderate Density Parity Check) codes. The main parameters are three integers (r, w, t) such that r is prime and 2 is a primitive root modulo r (the codelength is $n = 2r$), with $w \approx t \approx \sqrt{n}$ and $d = w/2$ odd.

- $\mathcal{H}_w = \{(h_0, h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = d = w/2\}$ the secret key space
- $\mathcal{E}_t = \{(e_0, e_1) \in \mathcal{R}^2 \mid |e_0| + |e_1| = t\}$ the error (or message) space
- $\Psi : \mathcal{R}^3 \rightarrow \mathcal{R}$ a QC-MDPC decoder
[$\Psi(e_0 h_0 + e_1 h_1, h_0, h_1) = (e_0, e_1)$ with probability close to 1 if $(h_0, h_1) \in \mathcal{H}_w$ and $(e_0, e_1) \in \mathcal{E}_t$ with appropriately selected parameters (r, w, t)]
- $(h_0, h_1) \in \mathcal{H}_w$ a secret key and $h = h_1 h_0^{-1} \in \mathcal{R}$ its associated public key, encryption and decryption are defined as follows:

$$\begin{array}{l|l} \text{Enc : } \mathcal{E}_t \times \mathcal{R} \rightarrow \mathcal{R} & \text{Dec : } \mathcal{R} \times \mathcal{H}_w \rightarrow \mathcal{E}_t \\ (e_0, e_1), h \mapsto e_0 + e_1 h & c, (h_0, h_1) \mapsto \Psi(ch_0, h_0, h_1) \end{array}$$

In the attack model that we will consider, the adversary knows the sparse polynomials $(e_0, e_1) \in \mathcal{E}_t$ and seeks the sparse secret polynomials $(h_0, h_1) \in \mathcal{H}_w$ by using information about the Hamming weight of the syndrome $e_0 h_0 + e_1 h_1$ which may leak from the decoding process $\Psi(e_0 h_0 + e_1 h_1, h_0, h_1)$. This may reveal the distance spectrum of h_0 or h_1 and allow the recovery of the secrets.

level	r	w	t
BIKE-1	12 323	142	134
BIKE-3	24 659	206	199
BIKE-5	40 973	274	264

Table 1. BIKE Parameters

HQC. In HQC, the decoding is not of the same nature. Still, in a known plaintext attack model, the adversary knows some sparse polynomials $(e, r_1, r_2) \in \mathcal{R}^3$ and seeks the sparse secret polynomials $(x, y) \in \mathcal{R}^2$ using information leakage which may occur during the decoding process $\text{Decode}(m\mathbf{G} + e')$, where $e' = e + r_1 y + r_2 x$. The situation is not identical, however we will see that if

information about the weight of e' leaks, then this may reveal the distance spectrum of x or y and allow the recovery of the secrets. Parameters are comparable to BIKE, for instance, for HQC-1, we have $r = 17669$, $|x| = |y| = 66$, which compares to $w/2$, and $|r_1| = |r_2| = |e| = 75$, which compares to $t/2$.

3 Key Reconstruction from a Full Spectrum

This section presents an algorithm to reconstruct the key from a full spectrum. It is inspired from [4] which is the first of its kind in the context of code-based cryptography. Starting from a distance spectrum S , it essentially consists in building an appropriate increasing sequence $f_0 \subset f_1 \subset \dots \subset f_\ell$ of polynomials, defined below as an S -reconstruction sequence, which eventually converges to the solution.

Definition 2. Let $S \in \mathbb{N}^{\lfloor \frac{r}{2} \rfloor}$, an S -reconstruction sequence is an increasing sequence $f_0 \subset f_1 \subset \dots \subset f_\ell$ of polynomials in $\mathcal{R}$ such that $\text{Sp}(f_\ell) \subset S$. It is valid if there exists $g \in \mathcal{R}$ such that $S = \text{Sp}(g)$ and $f_\ell \subset g$.

The algorithm recursively builds an S -reconstruction sequence and attempts to lengthen it in each possible way. It exits successfully when it finds $\text{Sp}(f_\ell) = S$ and backtracks when the sequence cannot be lengthened.

3.1 The S -extension

Definition 3 (S -extension). For any $S \in \mathbb{N}^{\lfloor r/2 \rfloor}$, the S -extension of $h \in \mathcal{R}$ is defined as

$$h^{(S)} = \sum_{i \in \mathcal{I}} x^i \text{ where } \mathcal{I} = \{i \mid 0 \leq i < r, \text{Sp}(h \cup x^i) \subset S\}.$$

For any h such that $\text{Sp}(h) \subset S$, its extension will contain any monomial that can be added to h and keep a spectrum included in S .

Lemma 1. For all $g, h \in \mathcal{R}$ and $S \in \mathbb{N}^{\lfloor r/2 \rfloor}$, we have

- (i) $h \subset g \Rightarrow g^{(S)} \subset h^{(S)}$
- (ii) $S = \text{Sp}(g) \Rightarrow g = g^{(S)}$

Proof. Let $h \subset g$ and $i \in g^{(S)}$. We have $\text{Sp}(h \cup x^i) \subset \text{Sp}(g \cup x^i)$ and $\text{Sp}(g \cup x^i) \subset S$, thus $i \in h^{(S)}$. Let $S = \text{Sp}(g)$, if $i \in g$ then $g = g \cup x^i$ and $i \in g^{(S)}$. Because of the strict monotony of the spectrum, if $i \notin g$ then $g \subsetneq g \cup x^i$ and $S \subsetneq \text{Sp}(g \cup x^i)$, thus $i \notin g^{(S)}$. $\square$

Lemma 1 implies the following proposition.

Proposition 2. Let $S \in \mathbb{N}^{\lfloor r/2 \rfloor}$, we consider a valid S -reconstruction sequence $f_0 \subset f_1 \subset \dots \subset f_\ell$. We have

$$\text{Sp}(f_\ell) \subset S \subset \text{Sp}(f_\ell^{(S)}) \text{ and } f_0^{(S)} \supset f_1^{(S)} \supset \dots \supset f_\ell^{(S)}$$

and there exists $g \in \mathcal{R}$ such that $S = \text{Sp}(g)$ and $f_\ell \subset g \subset f_\ell^{(S)}$.

In the execution of the algorithm, if the reconstruction sequence $f_0 \subset f_1 \subset \dots \subset f_\ell$ is valid at depth ℓ and grows towards a solution g , then the sequence of extensions is decreasing towards g and we have $f_\ell \subset g \subset f_\ell^{(S)}$. If the weight d of g is known, the reconstruction is complete if either $|f_\ell| = d$ or $|f_\ell^{(S)}| = d$.

For a valid reconstruction sequence, we observed that the weight of the extension decreases very quickly and provides a very efficient stopping condition.

ℓ	$\Pr[f_\ell^{(S)} = d]$	avg. $ f_\ell^{(S)} $
1	0	1422.13
2	0	253.68
3	0	99.23
4	0.0488	75.83
5	0.4953	71.93
6	0.8446	71.19
7	0.9599	71.04
8	0.9914	71.01
9	0.9971	71.00
10	0.9995	71.00
11	0.9999	71.00
12	1	71

Table 2. Statistics for reconstruction sequences $(f_\ell)_{\ell \geq 0}$ for 10 000 executions of Algorithm 1 with BIKE-1 parameters $(r, d) = (12\,323, 71)$. To get consistent statistics, we set $f_i^{(S)} = f_\ell^{(S)}$ for $i > \ell$ such that $|f_\ell^{(S)}| = d$ (though such f_i are never computed).

For instance, for BIKE-1, as shown in Table 2, the extension in Algorithm 1 will provide a solution, i.e. $|f_\ell^{(S)}| = d = 71$, at depth typically 5 or 6 and never more than 12, whereas the Hamming weight of f_ℓ is at most 2ℓ at depth ℓ .

3.2 Reconstruction Algorithm

Given a distance spectrum $S = (S_\delta)_{1 \leq \delta \leq r/2} \in \mathbb{N}^{\lfloor \frac{r}{2} \rfloor}$, Algorithm 1 seeks a polynomial $h \in \mathcal{R}$ of weight d such that $\text{Sp}(h) = S$. For every instance S and prior to the root recursive call, a set $\Delta = (\delta_i)_{i \geq 0}$ of distances belonging to the spectrum must be selected. The step-by-step rationale is the following:

Choosing Δ . The sequence Δ contains only values of δ such that $S_\delta > 0$ in any order. A given δ cannot appear more than S_δ times in the sequence. It is heuristically better to put the (first occurrence of) distances with the highest multiplicity at the beginning of the sequence Δ .

Root Call. The first step, depth $\ell = 0$, can be made simpler. Because of the spectrum's invariance under rotation, the first element in the reconstruction sequence can always be $1 + x^{\delta_0}$. In fact the root call of the recursive search will be $\text{KeySearchFull}(\ell = 1, g = 1 + x^{\delta_0}, S, \Delta)$. Hence, if we refer to a reconstruction sequence as in Definition 3, we have $f_0 = 0$ and $f_1 = 1 + x^{\delta_0}$.

Adding Monomials in Pairs. At depth ℓ , the input polynomial $g = f_\ell$ is augmented by including two monomials $x^s + x^{s+\delta_\ell}$ for all $0 \leq s < r$. Only the polynomials $f_{\ell+1} = f_\ell \cup (x^s + x^{s+\delta_\ell})$ such that $(f_i)_{0 \leq i \leq \ell+1}$ remains an S -reconstruction sequence are kept in $\mathcal{H}$. In other words, the polynomials $f_{\ell+1} = f_\ell \cup (x^s + x^{s+\delta_\ell})$ such that $\text{Sp}(f_{\ell+1}) \notin S$ are filtered out.

Stopping Conditions. Invalid reconstruction sequences will quickly become impossible to lengthen ($\mathcal{H} = \emptyset$), while valid reconstruction sequences will end up with $|f_\ell^{(S)}| = d$, the weight of the target solution.

Algorithm 1 Reconstruction From Full Spectrum

KeySearchFull : $(\ell, g, S, \Delta) \mapsto h$

Input: $\ell \geq 0$, $g \in \mathcal{R}$, $S \in \mathbb{N}^{\lfloor \frac{r}{2} \rfloor}$, $\Delta = (\delta_i)_{i \geq 0}$ with $S_{\delta_i} > 0$ for all $i \geq 0$

Output: $\perp$ or $h \in \mathcal{R}$ of Hamming weight d such that $\text{Sp}(h) = S$

- 1: $\mathcal{H} \leftarrow \{h = g \cup (x^s + x^{s+\delta_\ell}) \mid 0 \leq s < r, \text{Sp}(h) \subset S \subset \text{Sp}(h^{(S)}), |h^{(S)}| \geq d\}$
 - 2: **for** $h \in \mathcal{H}$ **do**
 - 3: **if** $|h^{(S)}| = d$ **then return** $h^{(S)}$
 - 4: $f \leftarrow \text{KeySearchFull}(\ell + 1, h, S, \Delta)$
 - 5: **if** $f \neq \perp$ **then return** f
 - 6: **return** $\perp$
-

Remark 1. a. In the worst case the Algorithm 1 has an exponential complexity.

In fact known algorithms in other fields for the “turnpike” or “beltway” problem are known to have worst case exponential time behavior [10]. It happens to be easy here, when the vector is sparse enough, but proving that the algorithm always succeeds early in that case seems challenging. At the moment, we simply give empirical evidence that the algorithm almost always succeeds in limited running time for parameters of interest.

- b. Algorithm 1 can be used without multiplicities, i.e. $S_\delta \in \{0, 1\}$.
- c. The algorithm returns a polynomial whose spectrum is equal to S . Because of the spectrum invariance under rotation and by mirroring, and depending on the context of the attack, checking and rotating or reversing might be necessary.

3.3 Results and Comparison

For BIKE-1 parameters, $(r, d, t) = (12323, 71, 134)$, our implementation of Algorithm 1 recovers h_0 from its full spectrum in about 2 milliseconds (on average over 10 000 independent executions). Using the syndrome weight as observable, the model of §3 and our experiments agree, and we get a faultless full spectrum 50% of the time with $2 \cdot 10^6$ queries, and 95% of the time with $2.7 \cdot 10^6$ queries.

We give in Section 7 more details about our experiments for BIKE, for other observables and for other schemes.

The primary purpose of the algorithm presented in this section was not to benchmark our solution, but to serve as a building block for the partial/soft information variants. In particular, we did not implement the other techniques, and we can only compare our performance to what was reported, often with incomplete information about platforms and/or programming languages. That being said, we measured performances that are orders of magnitude faster, sometimes by a factor 1000 to 10000, than performance claims from previous works with the possible exception of [5] in which a variant with similar features and performance is mentioned, but not described explicitly.

4 Reconstruction from a Partial Spectrum

In a practical scenario the full spectrum might not be available. Typically, the least reliable distances cannot be confidently classified as inside or outside the spectrum, and the reconstruction strategy must be adapted. In [9] partial information is handled by design, and the algorithm in [8] is adapted to partial information in [7].

4.1 Reconstruction Algorithm

For a given $S \in \mathbb{N}^{\lfloor \frac{r}{2} \rfloor}$, we assume that we know two tuples S^{lo} and S^{hi} in $\mathbb{N}^{\lfloor \frac{r}{2} \rfloor}$ such that $S^{\text{lo}} \subset S \subset S^{\text{hi}}$.

Rationale: Algorithm 1 gradually builds S -reconstruction sequences $(f_\ell)_{\ell \geq 0}$ and their S -extension $(f_\ell^{(S)})_{\ell \geq 0}$. A sequence is valid at step ℓ if $S \subset \text{Sp}(f_\ell^{(S)})$. If only bounds $S^{\text{lo}} \subset S \subset S^{\text{hi}}$ are available, the reconstruction is adapted as follows:

Enlarged Search Space. Any S -reconstruction sequence is also a S^{hi} -reconstruction sequence, thus if we explore all S^{hi} -reconstruction sequences we will explore in particular all S -reconstruction sequences. We explore more sequences, but we will eventually explore those that provide the solutions.

Weaker Pruning Criterion. The condition $S^{\text{hi}} \subset \text{Sp}(h^{(S^{\text{hi}})})$ is too strong and may discard sequences that are valid for S (but invalid for S^{hi}). Instead the test $S^{\text{lo}} \subset \text{Sp}(h^{(S^{\text{hi}})})$ will never exclude a valid S -reconstruction sequence.

Hence the adapted Algorithm 2 will reach a complete valid reconstruction sequence and a solution if Δ is large enough. Note that in addition, the set Δ must only contain distances that are in the spectrum, i.e. elements of S^{lo} .

4.2 Results and Comparison

The use of a partial spectrum was already considered in previous works. The algorithm in [9] has a different logic and direct comparison is difficult. The authors claim a reduction of 20% for the number of queries. The algorithm in

Algorithm 2 Reconstruction From Partial Spectrum

KeySearchPartial : $(\ell, g, S^{\text{hi}}, S^{\text{lo}}, \Delta) \mapsto h$

Input: $\ell \geq 0, g \in \mathcal{R}, S^{\text{hi}}, S^{\text{lo}} \subset \{1, \dots, \lfloor r/2 \rfloor\}, \Delta = (\delta_i)_{i \geq 0}$ with $S^{\text{lo}}_{\delta_i} > 0$ for all $i \geq 0$

Output: $\perp$ or $h \in \mathcal{R}$ such that $|h| = d$ and $S^{\text{lo}} \subset \text{Sp}(h) \subset S^{\text{hi}}$

```

1:  $\mathcal{H} \leftarrow \{h = g \cup (x^s + x^{s+\delta_\ell}) \mid 0 \leq s < r, \text{Sp}(h) \subset S^{\text{hi}}, S^{\text{lo}} \subset \text{Sp}(h^{(S^{\text{hi}})}), |h^{(S^{\text{hi}})}| \geq d\}$ 
2: for  $h \in \mathcal{H}$  do
3:   if  $|h^{(S^{\text{hi}})}| = d$  then return  $h^{(S^{\text{hi}})}$ 
4:    $f \leftarrow \text{KeySearchPartial}(\ell + 1, h, S^{\text{hi}}, S^{\text{lo}}, \Delta)$ 
5:   if  $f \neq \perp$  then return  $f$ 
6: return  $\perp$ 

```

[8] is comparable to ours, the authors measure the efficiency of their technique in terms of the proportion of unreliable distances (referred to as “suspicious”). Comparison is not straightforward but the order of magnitude for the amount of data needed to achieve the reconstruction seems to be the same. The algorithmic complexity is higher as for the full spectrum case.

Key Features:

Impact of the Tightness of the Bounds. Our experiments have highlighted the fact that a tight upper bound S^{hi} is much more important than a tight lower bound S^{lo} . Indeed, a loose upper bound may dramatically increase the number of paths that are explored recursively in the search space, while a loose lower bound will only delay the point when the reconstruction sequence is found invalid, without necessarily blowing up the number of paths in the search tree. For BIKE Level 1, the lower bound S^{lo} can be as low as 1% of the whole spectrum, whereas the running time blows up when the upper bound size exceeds 160% of the spectrum size, as shown in Table 3. Note also that with an exact upper bound the average running time with a 1% lower bound is only 4 ms instead of 2 ms with an exact lower bound.

$ S^{\text{hi}} / S $	100%	120%	130%	140%	150%	160%
Average running time (s)	0.004	0.03	0.1	0.37	2.2	13.13

Table 3. Average running time (in seconds) measured over 1 000 executions of Algorithm 2 for BIKE-1 parameters $(r, d) = (12323, 71)$ depending on the size of S^{hi} . The size of S^{lo} is fixed to 1% of $|S|$.

Connection with the Model. Properly selecting S^{lo} and S^{hi} from the observable obeys simple general rules but the process needs to be empirically adjusted for each set of parameters. We want to determine the smallest size for the sample set $\mathcal{E}$ such that there exists an upper limit L for $F_\delta(\mathcal{E}, (h_0, h_1))$ (see Figure 2) which verifies simultaneously:

- (1) the upper bound $S^{\text{hi}} = \{\delta \mid F_\delta \leq L\}$ is small enough, e.g. from Table 3 not more than $1.6 \cdot |S|$ for BIKE-1,
- (2) the probability $p = \Pr[\text{Sp}(h_0) \subset S^{\text{hi}}]$ to have the whole spectrum of the target secret below L is high enough.

As mentioned above, the lower limit for F_δ is never an issue and can be set relatively low. Using the model of §6 for the syndrome weight observable and for BIKE-1 parameters, we need $|\mathcal{E}| \approx 550\,000$ for a success probability $p = 0.5$ and $|\mathcal{E}| \approx 750\,000$ for a success probability $p = 0.95$. Those numbers match with our experiments and are less than a third of what is needed to obtain a full spectrum.

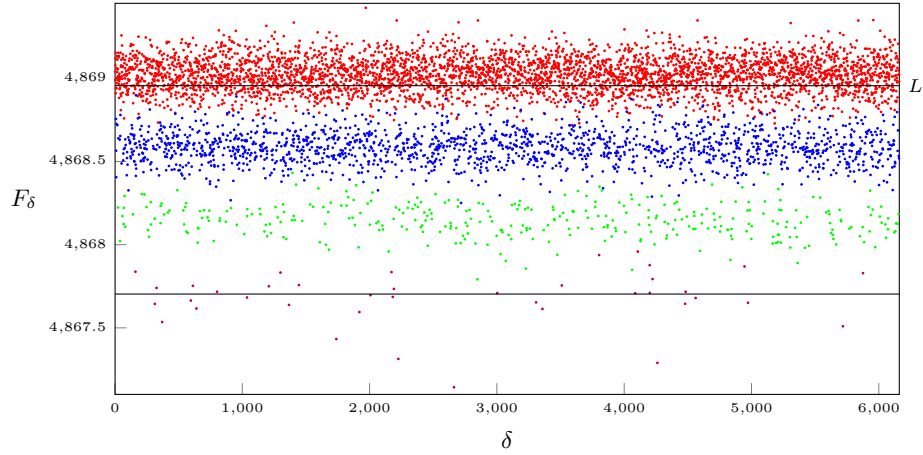


Fig. 2. $F_\delta(\mathcal{E}, (h_0, h_1))$ with $f(e_0, h_0) = |e_0 h_0 + e_1 h_1|$ and $|\mathcal{E}| = 7 \cdot 10^5$ for all distances δ for BIKE-1 parameters. The color of the distances indicate their multiplicity in the key spectrum (red = 0, blue = 1, green = 2, purple ≥ 3). L is the upper limit for $F_\delta(\mathcal{E}, (h_0, h_1))$ when using a partial spectrum.

5 Key Recovery from a Soft Distance Spectrum

When the number of queries reduces and there are too many unreliable distances, the partial spectrum approach fails. We either have too few data and the search becomes intractable or we have faults in the data and the algorithm fails.

To address this problem, we keep all the data and assign to each distance δ a probability distribution over all possible multiplicities, based on the observation.

5.1 Handling Soft Information

We define a soft distance spectrum as a tuple of probability distributions

$$S = (S_{\delta, \nu})_{1 \leq \delta \leq r/2, \nu \geq 0}. \quad (3)$$

Such a tuple can be obtained by modeling the observation as explained in Definition 6 of §6. For each distance δ , the sequence $(S_{\delta,\nu})_{\nu \geq 0}$ is a probability distribution with $S_{\delta,\nu}$ the probability to have $\nu = \mu_{h_0}(\delta)$, where h_0 is the target sparse secret polynomial of Hamming weight d .

Definition 4 (Score Function). *Given $S = (S_{\delta,\nu})_{1 \leq \delta \leq r/2, \nu \geq 0}$ a soft distance spectrum and a polynomial $h \in \mathcal{R}$ we define the score to be*

$$\text{score}(h, S) = \sum_{1 \leq \delta \leq r/2} \log \left(\sum_{\nu \geq \mu_h(\delta)} S_{\delta,\nu} \right). \quad (4)$$

The score of (h, S) essentially measures the (log of the) probability for a polynomial h to have its distance spectrum smaller than the distance spectrum of the secret polynomial whose observation led to the soft spectrum S . It is a negative number, and the highest values, i.e. closest to zero, are obtained for polynomials that have the highest probability to be included in a solution.

Definition 5 (Soft Extension). *Given $S = (S_{\delta,\nu})_{0 < \delta < r/2, \nu \geq 0}$ a soft distance spectrum and a polynomial $h \in \mathcal{R}$ we define a soft S -extension of h to be a tuple of all monomials x^i , $0 \leq i < r$ sorted in decreasing order of $\text{score}(h \cup x^i, S)$, that is*

$$h^{(S)} = (x^{j_0}, x^{j_1}, \dots, x^{j_{r-1}})$$

where $\text{score}(h \cup x^{j_i}, S) \geq \text{score}(h \cup x^{j_{i+1}}, S)$ for all $i = 0, \dots, r-2$. We will denote $h^{(S)}[\ell] = x^{j_\ell}$ and $h^{(S)}[0:d] = \sum_{0 \leq \ell < d} h^{(S)}[\ell] = x^{j_0} + \dots + x^{j_{d-1}}$.

5.2 Reconstruction Algorithm

Rationale: As in Algorithm 1, the reconstruction will recursively explore a reconstruction sequence $f_0 \subset f_1 \subset \dots \subset f_\ell$ and try to lengthen it in all plausible ways. The soft extension is used, as the extension in Algorithm 1, to extract the solution at an early stage. Hopefully, as when the extension has weight d in Algorithm 1, when we hit a solution, the first d monomials of the soft extension correspond to scores that are clearly better than the last $r-d$, and the solution is formed by those d monomials.

Step by step:

- Choice of Δ . It is best to use the distances that have the highest chances of belonging to the spectrum. We place in Δ the distances δ in decreasing order of the expected multiplicity $\sum_{\nu} \nu S_{\delta,\nu}$.
- At each recursive call of depth ℓ , the input polynomial g is augmented into h by adding (at most) two monomials $x^s + x^{s+\delta_\ell}$. We keep in $\mathcal{H}$ the values of h that have a score higher than a threshold T_ℓ which depends on ℓ and is determined empirically.

- For each $h \in \mathcal{H}$, we compute its soft S -extension: a score for each monomial x^i , $0 \leq i < r$, reflecting how likely is $h \cup x^i$ to belong to a solution given the soft spectrum S . We explore first the h that have the best d scores, in fact we look at the highest $\text{score}(h \cup x^i, S)$ where $x^i = h^{(S)}[d-1]$ is the d -th best monomial for h .
- If $h^{(S)}[0:d]$, the sum of the best d monomials for h , is not a solution, the recursive search continues at depth $\ell + 1$ with input h .

The algorithm is depth-first search. Pruning will happen when $\mathcal{H} = \emptyset$. The choice of the thresholds, or of any other rule to build the set of candidates $\mathcal{H}$, is empirical and must be fine-tuned for each set of parameters.

Algorithm 3 Reconstruction From Soft Spectrum

KeySearchSoft : $(\ell, g, S, \Delta) \mapsto h$

Input: $\ell \geq 0, g \in \mathcal{R}, S, \Delta = (\delta_i)_{i \geq 0}$

Output: $\perp$ or $h \in \mathcal{R}$ such that $|h| = d$

- 1: $\mathcal{H} \leftarrow \{h = g \cup (x^s + x^{s+\delta_\ell}), 0 \leq s < r, \text{score}(h, S) > T_\ell\}$
 $\triangleright (T_\ell)_{\ell \geq 0}$ determined empirically
 - 2: **for** $h \in \mathcal{H}$ in decreasing order of $\text{score}(h + h^{(S)}[d-1], S)$ **do**
 - 3: **if** “ $h^{(S)}[0:d]$ is good” **then return** $h^{(S)}[0:d]$
 - 4: $f \leftarrow \text{KeySearchSoft}(\ell + 1, h, S, \Delta)$
 - 5: **if** $f \neq \perp$ **then return** f
 - 6: **return** $\perp$
-

The assertion “ $h^{(S)}[0:d]$ is good” is admittedly a bit vague. In practice, in our cryptographic context, it is always possible to test the solution (and its reverse) against the public key. But, since it would be expensive to do that for all elements of $\mathcal{H}$, an efficient filter consists in looking at the d -th and the $(d+1)$ -th best scores in the soft extension. If we do not see a significant drop then we continue the execution without performing the full test. In practice, this heuristic filter selects the valid solutions and only them.

Empirically, we found that it was more efficient to select for $\mathcal{H}$ the augmented polynomials with a score high enough, and to browse through $\mathcal{H}$ in an order depending on the score of the soft extension. As it is written, though it is unlikely, the algorithm may fail to find a solution even with unlimited time. A more elaborate logic might provably exhaust the search space, but we do not believe it could reduce the order of magnitude of the complexity or improve the success rate.

5.3 Results

To recover a full spectrum for BIKE-1, more than $2 \cdot 10^6$ queries are required, in order to perfectly distinguish between multiplicities (stripes in scatter plot). With a soft spectrum, the reconstruction succeeds with 200 000 queries, about one tenth of the amount needed to get the full spectrum.

6 Observables and Models

We consider a cryptographic protocol which involves an encryption scheme based on QC-MDPC codes (e.g. BIKE). For a given secret key, a pair of sparse polynomials, the adversary can make queries with known plaintexts and can observe a quantity, an *observable*, related to the distance spectrum of the sparse secret key. This observable can be the decoding failure rate [4], the timing [2], the syndrome weight [2], or, more generally, any quantity that can be observed from a side-channel.

Here, we will focus on the syndrome weight. As remarked in [4,2], this quantity influences the decoding failure rate and the timing of the bit-flipping decoder of QC-MDPC codes. Failure and timing strongly depend on the parameter choice and on the exact variant of the decoding algorithm, while in contrast, the syndrome is a fundamental quantity of the scheme that inevitably appears during the decoding process. Its Hamming weight is prone to leakage, independently of any choice of parameters and algorithms. One of the purposes of this work is to expose and understand the vulnerabilities stemming from that.

An additional benefit of the syndrome weight as an observable is that it can also be used for HQC. Though there is no QC-MDPC decoding in it, HQC features a fundamental quantity similar to the syndrome and which also depends on the plaintext and on sparse secrets. Experiments show that if this quantity can be observed, it reveals the distance spectrum of the secrets as for BIKE.

6.1 Attack Model

For a given secret key $\mathbf{sk} = (h_0, h_1) \in \mathcal{H}_w$ and a known error pattern $\mathbf{e} = (e_0, e_1) \in \mathcal{E}_t$, the adversary may query a decryption oracle and observe a randomized numerical value $f(e_0, h_0)$ which we refer to as an *observable*. The key property of an observable for our purpose is that the expectation of $\mu_{e_0}(\delta)f(e_0, h_0)$ when $\mathbf{e}$ is uniformly distributed among weight- t vectors differs with $\mu_{h_0}(\delta)$. Hence, after observing multiple queries for the same secret key and a sample set of errors $\mathcal{E} \subset \mathcal{E}_t$, the adversary computes for all δ , $1 \leq \delta \leq r/2$, the quantity

$$F_\delta(\mathcal{E}, (h_0, h_1)) = \frac{1}{\sum_{\mathbf{e} \in \mathcal{E}} \mu_{e_0}(\delta)} \sum_{\mathbf{e} \in \mathcal{E}} \mu_{e_0}(\delta) f(e_0, h_0) \quad (5)$$

and guesses $\mu_{h_0}(\delta)$ from F_δ .

In this work, we will favor a scenario in which the adversary strictly follows the protocol, i.e. queries are made with uniformly distributed errors of Hamming weight t .

6.2 Model for BIKE's Syndrome Weight

The following proposition is proven in the appendix.

Proposition 3. For all $(h_0, h_1) \in \mathcal{H}_w$ and all δ , $1 \leq \delta \leq r/2$, we have

$$\frac{\sum_{(e_0, e_1) \in \mathcal{E}_t} \mu_{e_0}(\delta) |e_0 h_0 + e_1 h_1|}{\sum_{(e_0, e_1) \in \mathcal{E}_t} \mu_{e_0}(\delta)} = W_{\mu_{h_0}(\delta)}(n, w, t), \quad (6)$$

where for all integer $\nu \geq 0$

$$W_\nu(n, w, t) = (n/2 - w + \nu) \cdot S(n - 2, w, t - 2) + \nu \cdot S(n - 2, w - 2, t - 2) \\ + (w - 2\nu) \cdot (1 - S(n - 2, w - 1, t - 2)), \quad (7)$$

with

$$S(n, w, t) = \sum_{i \text{ odd}} \frac{\binom{w}{i} \binom{n-w}{t-i}}{\binom{n}{t}}.$$

- Remark 2.* a. Each value of W_ν corresponds to a stripe in the observation diagram or to a bump in the measured density (see Figure 1 for instance). Note that the difference $W_\nu - W_{\nu+1}$ is independent of ν .
- b. A formula similar to (6) is given in [2] and [5]. It measures an average syndrome weight conditioned by $\nu = \mu_{h_0}(\delta)$ and gives an approximation rather than an identity as we have here. This is discussed further later in this section.
- c. The proportion $\rho_\nu(r, d)$ of distances δ , $1 \leq \delta \leq r/2$, that appear in the spectrum of an element of $\mathcal{R}$ of weight d with multiplicity exactly ν is given in [11] (given here in Proposition 1). Together with the above proposition those numbers will characterize the model for the density of F_δ . For instance for BIKE-1

ν	W_ν	ρ_ν
0	4869.007	0.667
1	4868.573	0.272
2	4868.139	0.054
3	4867.705	0.007

Modeling with a Combination of Normal Distributions. From Proposition 3, for any (h_0, h_1) and any δ , the average value of $F_\delta(\mathcal{E}, (h_0, h_1))$ as defined in (5) with $f(e_0, h_0) = |e_0 h_0 + e_1 h_1|$ over all possible choices of $\mathcal{E} \subset \mathcal{E}_t$ is equal to $W_\nu(n, w, t)$ where $\nu = \mu_{h_0}(\delta)$. From our observation, which is consistent with other works, e.g. [5,7], the probability $\Pr[F_\delta = x \mid \nu = \mu_{h_0}(\delta)]$ is accurately modeled by a normal distribution $\mathcal{N}(W_\nu, \sigma^2)$. The mean⁵ W_ν is given by Proposition 3. The standard deviation σ is determined empirically and we observed in our experiments that, independently of ν , it is close to $\sigma_0/\sqrt{N}$ where $N = |\mathcal{E}|$

⁵ here and in similar situations throughout the paper we dropped the arguments (n, w, t) as there is no ambiguity in the context

and σ_0 only depends on the system parameters (n, r, w, d, t) . The probability density function of $F_\delta(\mathcal{E}, (h_0, h_1))$ for a given $\nu = \mu_{h_0}(\delta)$ is well modeled by

$$\varphi_\nu(x) = \frac{\sqrt{|\mathcal{E}|}}{\sigma_0 \sqrt{2\pi}} \exp\left(-\frac{|\mathcal{E}|}{2\sigma_0^2}(x - W_\nu)^2\right), \quad (8)$$

and, denoting by ρ_ν the proportion of distances of multiplicity ν given in Proposition 1, we get the following model for the probability distribution function of $F_\delta(\mathcal{E}, (h_0, h_1))$

$$\varphi(x) = \sum_{\nu \geq 0} \rho_\nu \varphi_\delta(x). \quad (9)$$

We observe in Figure 3, for $|\mathcal{E}| = 10^6$ and BIKE-1 parameters, that the model in dashed blue is relatively close to the measured density in red.

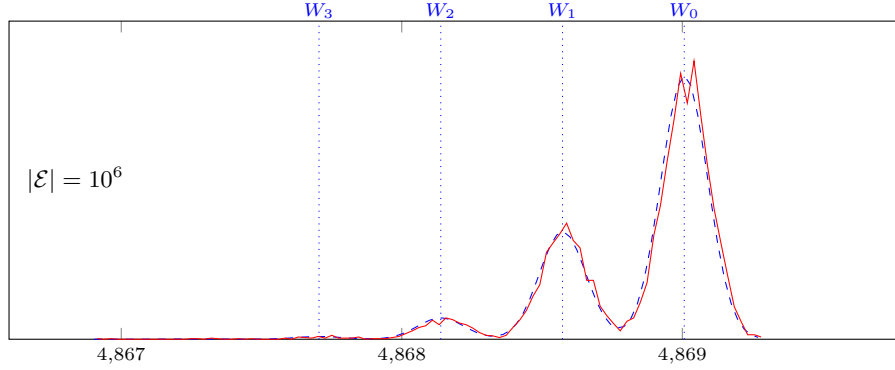


Fig. 3. Density of $F_\delta(\mathcal{E}, (h_0, h_1))$ for all distances δ and for $(r, d, t) = (12323, 71, 134)$, $w = 142$, $n = 24646$. The observed density is in red and the model in dashed blue. The vertical dotted blue lines indicate the means $W_\nu(n, w, t)$ according to multiplicity $\nu = \mu_{h_0}(\delta)$.

Centering the Model and the Observation. For BIKE-1 parameters, the expected syndrome weight is equal to $r \cdot S(n, w, t) = 4868.831$ while for the error patterns in $\mathcal{E}$ used in Figure 3, the average is 4868.848. The difference is +0.017, small but measurable. If we compare W_ν with the measured average values of F_δ conditioned with ν we observe the same difference in the whole range:

ν	$W_\nu(n, w, t)$	avg. $F_\delta \nu$	difference
0	4869.007	4869.023	+0.016
1	4868.573	4868.588	+0.015
2	4868.139	4868.159	+0.020
3	4867.705	4867.726	+0.021

In fact, a close observation of Figure 3 reveals that the measured data, the red curve, is slightly shifted to the right compared to the model. This shift could go left as well. This variation is expected as the standard deviation of the syndrome weight is relatively high, e.g. ≈ 50 for BIKE-1 parameters, and for instance if $N = |\mathcal{E}| = 10^6$, the standard deviation of the average syndrome weight is still $\approx 50/\sqrt{N} = 0.05$ which is not so small compared to the difference $W_\nu - W_{\nu+1} \approx 0.434$ we are trying to detect.

This can be handled by centering. We subtract the average syndrome weight $r \cdot S(n, w, t)$ from the modeled distribution function input, and we subtract the observed average syndrome weight from F_δ . We get a centered model

$$\bar{\varphi}_\nu(x) = \varphi_\nu(x - r \cdot S(n, w, t)) \quad \text{and} \quad \bar{\varphi}(x) = \sum_{\nu \geq 0} \rho_\nu \bar{\varphi}_\nu(x) \quad (10)$$

and a centered observation

$$\bar{F}_\delta(\mathcal{E}, (h_0, h_1)) = F_\delta(\mathcal{E}, (h_0, h_1)) - \frac{1}{|\mathcal{E}|} \sum_{(e_0, e_1) \in \mathcal{E}} |e_0 h_0 + e_1 h_1| \quad (11)$$

which give a more accurate estimate of the multiplicities and their reliability.

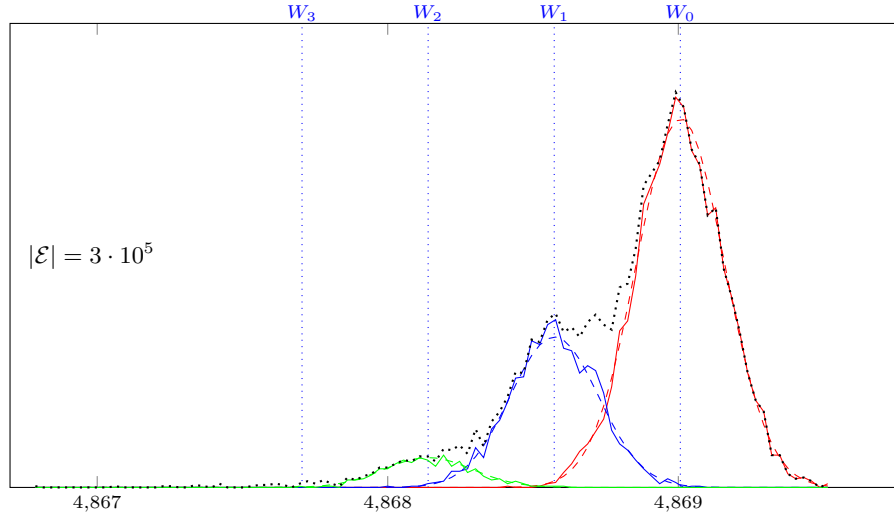


Fig. 4. Density of $\bar{F}_\delta(\mathcal{E}, (h_0, h_1))$ for all δ splitted according to $\nu = \mu_{h_b}(\delta)$ ($\nu = 0$ in red, $\nu = 1$ in blue, $\nu = 2$ in green, $\nu \geq 3$ not represented), the model $\rho_\nu \bar{\varphi}_\nu(x)$ appears in dashes. The cumulated value $\bar{F}_\delta$ is shown in dotted black. The vertical dotted blue lines indicate the means $W_\nu(n, w, t)$ according to multiplicity $\nu = \mu_{h_0}(\delta)$.

Soft Information from the Centered Average Syndrome Weight. To use the soft input algorithm of §5, we need to build a soft distance spectrum (3) from the observation. For each δ and each ν , according to the model, the probability to have $\mu_{h_0}(\delta) = \nu$ given the (centered) observation $\bar{F}_\delta$ is

$$\Pr [\mu_{h_0}(\delta) = \nu \mid \bar{F}_\delta = x] = \frac{\rho_\nu \cdot \bar{\varphi}_\nu(x)}{\bar{\varphi}(x)}.$$

Definition 6 (Soft Spectrum from Observation). *For a given secret key (h_0, h_1) , the observable $f(e_0, h_0) = |e_0 h_0 + e_1 h_1|$ is known for all (e_0, e_1) in the sample set $\mathcal{E} \subset \mathcal{E}_t$. The soft distance spectrum used in Algorithm 3 is defined for all δ , $1 \leq \delta \leq r/2$, and all $\nu \geq 0$ by*

$$S_{\delta, \nu} = \frac{\rho_\nu(r, d) \cdot \bar{\varphi}_\nu(\bar{F}_\delta(\mathcal{E}, (h_0, h_1)))}{\bar{\varphi}(\bar{F}_\delta(\mathcal{E}, (h_0, h_1)))}$$

where $\bar{F}_\delta$ is defined by (5) and (11), $\bar{\varphi}$ and $\bar{\varphi}_\nu$ are defined by (9), (8) and (10), and ρ_ν is defined in Proposition 1.

Comment About the Use of the Weighted Average. In our modeling, we have considered for F_δ the average of $f(e_0, h_0)$ weighted with the error multiplicity (5) rather than simply the average of $f(e_0, h_0)$ for error patterns such that $\delta \in \text{Sp}(e_0)$

$$F'_\delta = \frac{1}{|\mathcal{E}_\delta|} \sum_{e_0 \in \mathcal{E}_\delta} f(e_0, h_0) \text{ where } \mathcal{E}_\delta = \{e \in \mathcal{E} \mid \delta \in \text{Sp}(e_0)\}$$

as in previous works [2,5]. The main benefit is a larger difference between the multiplicities. For BIKE-1 parameters, we get

ν	avg. $F'_\delta \nu$	avg. $F_\delta \nu$	W_ν
0	4868.9778	4869.0072	4869.0076
1	4868.6144	4868.5704	4868.5733
2	4868.2496	4868.1360	4868.1391
3	4867.9006	4867.7081	4867.7049

The above table compares the average values of F'_δ and F_δ for a given ν . We observe a difference of 0.36 in the first column rather than 0.43 in the second. Thus F_δ will be better to discriminate between the multiplicities. Another benefit is the Proposition 3 which gives a closed formula (7) for W_ν which is met, on average, by F_δ . Previous works also give the formula (7) for the average syndrome weight but the small deviation with experiments was already observed [2].

7 Variants and Numerical Results

The framework we have described can be applied to other observables $f(e_0, h_0)$ for BIKE. It is relevant if the average value of $\mu_{e_0}(\delta) \cdot f(e_0, h_0)$ differs with the

multiplicity of δ in the spectrum of h_0 . The model, i.e. the (centered) probability density functions $\bar{\varphi}_\nu$ for $\nu \geq 0$, which can be established empirically by simulation if not available theoretically, will allow the construction of soft distance spectra to be used in Algorithm 3.

In this section, we first give numerical results for all security levels of BIKE when the observable is the syndrome weight. Then we will consider other relevant observables, timing or noisy syndrome weight, which are closer to the kind of leakage obtained through side channels. Finally, we show how the framework, observable and model, can be applied to recover sparse secret polynomials in HQC. The experiments were performed using the implementation described in Section 1.4. The code used to obtain the results in this section uses multithreading and is available at https://github.com/m-salom/key_rec_spectrum.

7.1 Syndrome Weight

We first summarize the results considering the Hamming weight of the syndrome as an observable, i.e. $f(e_0, h_0) = |e_0 h_0 + e_1 h_1|$, which was considered throughout this paper.

Full Spectrum. A full spectrum is obtained when there is a complete separation between distances with multiplicity 0 and 1. We want to determine the smallest size of $|\mathcal{E}|$ such that the full spectrum is exactly corresponding to the set of distances below some limit L with probability p . We set $L = \frac{W_0 + W_1}{2}$.

Using the model described in §6, Table 4 reports the minimal number of queries $|\mathcal{E}|$ to obtain a full spectrum with success probabilities $p = 0.5$ and $p = 0.95$, for different BIKE security levels.

	$p = 0.5$	$p = 0.95$
BIKE-1	≈ 2	≈ 2.7
BIKE-3	≈ 5.2	≈ 7
BIKE-5	≈ 11	≈ 14.6

Table 4. Minimal size of $\mathcal{E}$ (rounded to the nearest 100 000) in millions of queries to have a full spectrum with probability p for 3 levels of BIKE.

Soft Spectrum. The soft spectrum can be obtained using the process described in Definition 6. Table 5 gives the number of queries and average running time required to successfully recover the key using soft information in Algorithm 3 on BIKE-1 parameters. The algorithm always terminates but sometimes without recovering the key. The average running time is computed over successful runs. We achieve the reconstruction of the key 50% of the time with 200 000 queries, about one tenth of the amount needed to get the full spectrum.

	$p = 0.5$	$p = 0.95$
$ \mathcal{E} $	200 000	300 000
Avg running time (s)	155	70

Table 5. Average running time (in seconds, rounded up) measured over 100 executions for size of $\mathcal{E}$ chosen near the minimal size required to recover the key using Algorithm 3 with probability p for BIKE-1 parameters.

7.2 Noisy Syndrome Weight

In a practical side-channel attack, an attacker may have access to a noisy version of the syndrome weight. To better reflect a realistic scenario, we extend the framework to noisy observations of the form:

$$f(e_0, h_0) = |e_0 h_0 + e_1 h_1| + \mathcal{N}(0, \sigma^2)$$

where the noise is modeled with a Gaussian distribution of mean 0 and of standard deviation σ which is a common model in side-channel contexts.

Being more specific would require targeting a specific implementation, which is out of the scope of this paper. The point of this section is to establish that our work applies to any side-channel leaking information correlated to the syndrome weight.

Full Spectrum. The experiments were run for three levels of noise that were chosen arbitrarily for BIKE-1 parameters. Using the model of §6 adapted to take the noise into account, Table 6 shows the minimal number of queries needed to recover a full spectrum with different probabilities.

	$p = 0.5$	$p = 0.95$
$\sigma = 10$	≈ 2	≈ 2.7
$\sigma = 30$	≈ 2.2	≈ 3
$\sigma = 50$	≈ 2.7	≈ 3.6

Table 6. Minimal size of $\mathcal{E}$ (rounded to the nearest 100 000) in millions of queries to have a full spectrum with probability p for different levels of noise σ for BIKE-1 parameters.

Soft Spectrum. Increasing the noise level impacts the benefit of using soft information. As shown in Table 7, when the noise increases, the gain provided by soft information decreases from a factor 10 in the noiseless case to 6.75 for $\sigma = 50$.

	$ \mathcal{E} $	Avg running time (s)
$\sigma = 10$	200 000	205
$\sigma = 30$	280 000	259
$\sigma = 50$	400 000	304

Table 7. Average running time (in seconds, rounded up) measured over 100 executions for size of $\mathcal{E}$ chosen near the minimal size required to recover the key using Algorithm 3 with probability 0.5 for different levels of noise σ for BIKE-1 parameters.

7.3 Timing Attack

Another observable we considered is the number of iterations in the bit-flipping algorithm: $f(e_0, h_0) = i$. This quantity may be measured through a side-channel like timing and is correlated to the syndrome weight hence, it behaves the same way.

The main point of this section is to observe that the weighted average (i.e. weighting the average observation with the error multiplicity) is relevant for timing, as we believe it is for any observable in this framework. We did not consider soft information in this case, though it could be done after building an empirical observation model from simulations.

Full Spectrum. Following the initial scenario namely strictly following the protocol, for BIKE-1 parameters $(r, d, t) = (12323, 71, 134)$, the experiments show that this observable requires a significantly higher amount of queries than the syndrome weight, about 1000 times more. To recover a full spectrum, we need $|\mathcal{E}| \approx 2.2 \cdot 10^9$ for a success probability $p = 0.5$ and $|\mathcal{E}| \approx 3 \cdot 10^9$ for a success probability $p = 0.95$. Those numbers seem hard to reach in practice.

To deal with this issue, we can consider a scenario in which the error weight t can be increased. The experiments for BIKE-1 parameters with an increased error weight $(r, d, t) = (12323, 71, 180)$ show that more queries are still needed than with the syndrome weight, but the factor drops to 20 instead of 1000. We need $|\mathcal{E}| \approx 4 \cdot 10^7$ for a success probability $p = 0.5$ and $|\mathcal{E}| \approx 5.5 \cdot 10^7$ for a success probability $p = 0.95$ to recover a full spectrum.

Weighted Average. For this observable, we have again considered the average of $f(e_0, h_0)$ weighted with the multiplicity in the error. Just like for the syndrome weight, it allows a better discrimination between multiplicities:

ν	$(r, d, t) = (12323, 71, 134)$		$(r, d, t) = (12323, 71, 180)$	
	avg. $F'_\delta \nu$	avg. $F_\delta \nu$	avg. $F'_\delta \nu$	avg. $F_\delta \nu$
0	2.999572	2.9995727	8.884984	8.885543
1	2.999567	2.9995660	8.881557	8.880761
2	2.999562	2.9995602	8.877729	8.875403

For $(r, d, t) = (12323, 71, 134)$, we observe a difference of $5.59 \cdot 10^{-6}$ between multiplicity 0 and 1 in the first column while in the second column, the difference is $6.674 \cdot 10^{-6}$.

For $(r, d, t) = (12323, 71, 180)$, the average number of iterations is increased. We observe a difference of 0.00343 between multiplicity 0 and 1 in the first column and of 0.00478 in the second column.

7.4 HQC

We will use the notation of the HQC specification, except for the blocksize which we denote r instead of n in [3]. Let $\mathcal{R} = \mathbb{F}_2[x]/(x^r - 1)$, for any integer ω , we will denote $\mathcal{R}_\omega = \{g \in \mathcal{R} \mid |g| = \omega\}$. HQC parameters are $(r, \omega, \omega_r, \omega_e)$ with $\omega \approx \omega_r \approx \omega_e \approx \sqrt{r}/2$. The scheme features a secret key $(x, y) \in \mathcal{R}_\omega^2$, and for each Encapsulation/Decapsulation instance a triple $(r_1, r_2, e) \in \mathcal{M} = \mathcal{R}_{\omega_r}^2 \times \mathcal{R}_{\omega_e}$ is selected uniformly at random. For HQC-1, $r = 17669, \omega = 66$ and $\omega_r = \omega_e = 75$. We assume that the adversary can observe the Hamming weight of $e' = r_1 y + r_2 x + e$ through a side channel and will try to discover the spectrum of x .

Proposition 4. *For all $(x, y) \in \mathcal{R}_\omega^2$ and all $\delta, 1 \leq \delta \leq r/2$, we have*

$$\frac{\sum_{(r_1, r_2, e) \in \mathcal{M}} \mu_{r_2}(\delta) |r_1 y + r_2 x + e|}{\sum_{(r_1, r_2, e) \in \mathcal{M}} \mu_{r_2}(\delta)} = W_{\mu_x(\delta)}^{\text{HQC}}(r, \omega, \omega_r, \omega_e), \quad (12)$$

where for all integer $\nu \geq 0$

$$W_\nu^{\text{HQC}}(r, \omega, \omega_r, \omega_e) = \omega' + (1 - 2\omega'/r) \cdot \omega_1 \quad (13)$$

with

$$\begin{aligned} \omega_1 = (r - 2\omega + \nu) \cdot S(r - 2, \omega, \omega_r - 2) + \nu \cdot S(r - 2, \omega, \omega_r - 2) \\ + 2(\omega - \nu) \cdot (1 - S(r - 2, \omega, \omega_r - 2)) \end{aligned} \quad (14)$$

and

$$\omega' = \omega_e + (r - 2\omega_e) \cdot S(r, \omega, \omega_r). \quad (15)$$

The proposition is proven in appendix. We remark that in (13), only ω_1 depends on the multiplicity ν , and this dependency is very similar to what we have for BIKE. Hence, after observation of a sample set $\mathcal{E} \subset \mathcal{M}$ drawn uniformly at random, we get

$$F_\delta(\mathcal{E}, (x, y)) = \frac{\sum_{(r_1, r_2, e) \in \mathcal{E}} \mu_{r_2}(\delta) |r_1 y + r_2 x + e|}{\sum_{(r_1, r_2, e) \in \mathcal{E}} \mu_{r_2}(\delta)}.$$

As in §6, we gain some accuracy by centering the observation and the model. The average weight of $|r_1 y + r_2 x + e|$ is equal to $\omega' + (1 - 2\omega'/r) \cdot S(r, \omega, \omega_r)$ where ω' is defined by (15). The approach used for BIKE in Definition 6 can be applied and provides a soft distance spectrum to be used in Algorithm 3.

Full Spectrum. Using the model described above, we recover a full spectrum with probability 0.5 using $1.2 \cdot 10^6$ queries and with probability 0.95 using $1.6 \cdot 10^6$ queries which matches our experiments.

Soft Spectrum. We can adapt the process in Definition 6 using HQC-1 parameters and model to obtain a soft spectrum. Table 8 shows that the numbers of queries to get a full spectrum are reduced by a factor of about 10, as for BIKE.

	$p = 0.5$	$p = 0.95$
$ \mathcal{E} $	110 000	190 000
Avg running time (s)	239	61

Table 8. Average running time (in seconds, rounded up) measured over 100 executions for size of $\mathcal{E}$ chosen near the minimal size required to recover the key using Algorithm 3 with probability p for HQC-1 parameters.

8 Conclusion

In this paper we have fully specified and implemented an algorithm to efficiently recover a sparse polynomial from its distance spectrum using soft information. For BIKE, we have shown that, in an idealized leakage model in which the adversary has access to the syndrome weight, the number of queries to recover the secret key is significantly reduced, by a factor 10 for BIKE-1. Our framework also applies to HQC if an adversary has access to the error weight, and the soft information approach also reduces the number of queries, by a factor 10 for HQC-1. Finally, we have given evidence that the framework also applies to more realistic leakage models, for instance a noisy observation of the syndrome weight in BIKE, with a significant reduction in the number of queries when soft information is used.

An important takeaway of this work for BIKE or HQC implementors, is to be aware of the need to avoid any leakage of the syndrome weight or the error weight in an implementation of those schemes.

Acknowledgment. This research was partially funded by the French Agence Nationale de la Recherche and plan France 2030 program under grant ANR-22-PETQ-0008 PQ-TLS.

References

1. Cieliebak, M., Eidenbenz, S.J., Penna, P.: Noisy data make the partial digest problem np-hard. In: Benson, G., Page, R.D.M. (eds.) Algorithms in Bioinformatics, Third International Workshop, WABI 2003, Budapest, Hungary, September 15-20,

- 2003, Proceedings. Lecture Notes in Computer Science, vol. 2812, pp. 111–123. Springer (2003). https://doi.org/10.1007/978-3-540-39763-2_9
2. Eaton, E., Lequesne, M., Parent, A., Sendrier, N.: QC-MDPC: A timing attack and a CCA2 KEM. In: Post-Quantum Cryptography - 9th International Conference, PQCrypto 2018, Fort Lauderdale, FL, USA, April 9–11, 2018, Proceedings. pp. 47–76 (2018), https://doi.org/10.1007/978-3-319-79063-3_3
 3. Gaborit, P., Aguilar-Melchor, C., Aragon, N., Bettaieb, S., Bidoux, L., Blazy, O., Deneuville, J.C., Persichetti, E., Zémor, G., Bos, J., Dion, A., Lacan, J., Robert, J.M., Véron, P., Barreto, P.L., Ghosh, S., Gueron, S., Güneysu, T., Misoczki, R., Richter-Brokmann, J., Sendrier, N., Tillich, J.P., Vasseur, V.: HQC (Aug 2025), <https://pqc-hqc.org>
 4. Guo, Q., Johansson, T., Stankovski, P.: A key recovery attack on MDPC with CCA security using decoding errors. In: Cheon, J.H., Takagi, T. (eds.) Advances in Cryptology - ASIACRYPT 2016. LNCS, vol. 10031, pp. 789–815 (2016). http://dx.doi.org/10.1007/978-3-662-53887-6_29
 5. Guo, Q., Johansson, T., Stankovski, P.: A key recovery reaction attack on QC-MDPC. IEEE Transactions on Information Theory **65**(3), 1845–1861 (2019). <https://doi.org/10.1109/TIT.2018.2877458>
 6. Huang, S., Dokmanic, I.: Reconstructing point sets from distance distributions. IEEE Trans. Signal Process. **69**, 1811–1827 (2021). <https://doi.org/10.1109/TSP.2021.3063458>
 7. Ohtsuka, M., Ishimaru, T., Iseki, R., Kukita, S., Watanabe, K.: Key reconstruction for QC-MDPC McEliece from imperfect distance spectrum. Cryptology ePrint Archive, Paper 2025/485 (2025), <https://eprint.iacr.org/2025/485>
 8. Ohtsuka, M., Ishimaru, T., Tsukie, Y., Kukita, S., Watanabe, K.: Efficient reconstruction in key recovery attack on the QC-MDPC McEliece cryptosystems. IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences **E108.A**(3), 414–422 (2025). <https://doi.org/10.1587/transfun.2024TAP0003>
 9. Paiva, T., Terada, R.: Improving the efficiency of a reaction attack on the QC-MDPC McEliece. IEICE Transactions on Fundamentals of Electronics Communications and Computer Sciences **E101.A**, 1676–1686 (10 2018). <https://doi.org/10.1587/transfun.E101.A.1676>
 10. Skiena, S., Smith, W.D., Lemke, P.: Reconstructing sets from interpoint distances (extended abstract). In: Seidel, R. (ed.) Proceedings of the Sixth Annual Symposium on Computational Geometry, Berkeley, CA, USA, June 6–8, 1990. pp. 332–339. ACM (1990). <https://doi.org/10.1145/98524.98598>
 11. Vasseur, V.: Post-quantum cryptography: a study of the decoding of QC-MDPC codes. Thesis, Université de Paris (Mar 2021), <https://tel.archives-ouvertes.fr/tel-03254461>

A Proof of Proposition 3

We denote $\mathcal{W}_{n,t}$ the set of words of weight t in $\mathbb{F}_2^n$. For any given $h \in \mathbb{F}_2^n$, we admit the following result

$$\mathbb{E} \left[\langle h, e \rangle = 1 \mid e \leftarrow^{\$} \mathcal{W}_{n,t} \right] = S(n, |h|, t) \quad (16)$$

where $\langle \mathbf{h}, \mathbf{e} \rangle$ denote the scalar product of $\mathbf{h}$ and $\mathbf{e}$ and for any integers n, w, t

$$S(n, w, t) = \sum_{i \text{ odd}} \frac{\binom{w}{i} \binom{n-w}{t-i}}{\binom{n}{t}}.$$

Lemma 2. For any matrix $\mathbf{H} = (h_{i,j})_{0 \leq i < r, 0 \leq j < n} \in \mathbb{F}_2^{r \times n}$ of constant row weight w ,

$$\mathbb{E} \left[|\mathbf{e} \mathbf{H}^\top| \mid \mathbf{e} \leftarrow^{\$} \mathcal{W}_{n,t} \right] = r \cdot S(n, w, t).$$

Proof. Let $\mathbf{h}_i$, $0 \leq i < r$ denote the i -th row of $\mathbf{H}$, we easily derives from (16):

$$\begin{aligned} \mathbb{E} \left[|\mathbf{e} \mathbf{H}^\top| \mid \mathbf{e} \leftarrow^{\$} \mathcal{W}_{n,t} \right] &= \frac{1}{\binom{n}{t}} \sum_{|\mathbf{e}|=t} |\mathbf{e} \mathbf{H}^\top| \\ &= \frac{1}{\binom{n}{t}} \sum_{|\mathbf{e}|=t} \sum_{0 \leq i < r} \langle \mathbf{h}_i, \mathbf{e} \rangle \\ &= \sum_{0 \leq i < r} \frac{1}{\binom{n}{t}} \sum_{|\mathbf{e}|=t} \langle \mathbf{h}_i, \mathbf{e} \rangle \\ &= \sum_{0 \leq i < r} S(n, |\mathbf{h}_i|, t) = r \cdot S(n, w, t). \end{aligned}$$

□

Note that the lemma holds for any matrix as long as the row weight condition is verified, in particular, a dependence between rows (e.g. as in block circulant matrices) does not affect the result. Dependence between rows would affect the standard deviation though.

Lemma 3. For any matrix $\mathbf{H} = (h_{i,j})_{0 \leq i < r, 0 \leq j < n} \in \mathbb{F}_2^{r \times n}$ of constant row weight w , and for any pair of distinct columns indices $0 \leq j, j' < n$, we have

$$\begin{aligned} \mathbb{E} \left[|\mathbf{e} \mathbf{H}^\top| \mid \mathbf{e} \leftarrow^{\$} \mathcal{W}_{n,t}, e_j = e_{j'} = 1 \right] &= \nu_0 \cdot S(n-2, w, t-2) \\ &\quad + \nu_1 \cdot (1 - S(n-2, w-1, t-2)) + \nu_2 \cdot S(n-2, w-2, t-2), \end{aligned} \quad (17)$$

where $\nu_\ell = |\{i, 0 \leq i < r \mid |(h_{i,j}, h_{i,j'})| = \ell\}|$, $0 \leq \ell < 2$.

Proof. We reorder the columns of $\mathbf{H}$ so that the j -th and j' -th columns appear left, and we reorder the rows of $\mathbf{H}$ in increasing order relatively the weight of those two coordinates. We get

	2	$n-2$	
$\mathbf{H}' =$	00	$\mathbf{H}'_0 \in \mathbb{F}_2^{\nu_0 \times (n-2)}$	ν_0
	01 10	$\mathbf{H}'_1 \in \mathbb{F}_2^{\nu_1 \times (n-2)}$	ν_1
	11	$\mathbf{H}'_2 \in \mathbb{F}_2^{\nu_2 \times (n-2)}$	ν_2

with $\mathbf{H}'_\ell$, $0 \leq \ell \leq 2$, of row weight $w - \ell$ respectively. For $\mathbf{e} \in \mathbb{F}_2^n$, we write $\mathbf{e} = (e_0, e_1, \mathbf{e}')$ with $\mathbf{e}' \in \mathbb{F}_2^{n-2}$. If $e_0 = e_1 = 1$, we have (with $\mathbf{1}$ the all one vector)

$$\mathbf{e}\mathbf{H}'^\top = (\mathbf{e}'\mathbf{H}'_0{}^\top, \mathbf{1} + \mathbf{e}'\mathbf{H}'_1{}^\top, \mathbf{e}'\mathbf{H}'_2{}^\top) \text{ and } \left| \mathbf{e}\mathbf{H}'^\top \right| = \left| \mathbf{e}'\mathbf{H}'_0{}^\top \right| + \nu_1 - \left| \mathbf{e}'\mathbf{H}'_1{}^\top \right| + \left| \mathbf{e}'\mathbf{H}'_2{}^\top \right|. \quad (18)$$

From Lemma 2, we get

$$\mathbb{E} \left[\left| \mathbf{e}'\mathbf{H}'_\ell{}^\top \right| \mid \mathbf{e}' \leftarrow^{\$} \mathcal{W}_{n-2, t-2} \right] = \nu_\ell \cdot S(n-2, w-\ell, t-2), 0 \leq \ell \leq 2. \quad (19)$$

The following identities

$$\begin{aligned} \mathbb{E} \left[\left| (1, 1, \mathbf{e}')\mathbf{H}'^\top \right| \mid \mathbf{e}' \leftarrow^{\$} \mathcal{W}_{n-2, t-2} \right] &= \mathbb{E} \left[\left| \mathbf{e}\mathbf{H}'^\top \right| \mid \mathbf{e} \leftarrow^{\$} \mathcal{W}_{n, t}, e_0 = e_1 = 1 \right] \\ &= \mathbb{E} \left[\left| \mathbf{e}\mathbf{H}^\top \right| \mid \mathbf{e} \leftarrow^{\$} \mathcal{W}_{n, t}, e_j = e_{j'} = 1 \right]. \end{aligned}$$

together with (18) and (19) give (17), which completes the proof. $\square$

Lemma 4. *Let $r = n/2$, $(h_0, h_1) \in \mathcal{H}_w$, $b \in \{0, 1\}$ and $1 \leq \delta \leq \lfloor r/2 \rfloor$. For all i , $0 \leq i < r$, define $\mathcal{E}_t(b, \delta, i) = \{(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t \mid e_{b,i} = e_{b,i+\delta} = 1\}$. We have*

$$\begin{aligned} \text{(i)} \quad & \sum_{0 \leq i < r} |\mathcal{E}_t(b, \delta, i)| = \sum_{(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t} \mu_{\mathbf{e}_b}(\delta), \\ \text{(ii)} \quad & \sum_{0 \leq i < r} \sum_{(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t(b, \delta, i)} |e_0 h_0 + e_1 h_1| = \sum_{(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t} \mu_{\mathbf{e}_b}(\delta) |e_0 h_0 + e_1 h_1|, \\ \text{(iii)} \quad & \text{for all } i, 0 \leq i < r, \quad \sum_{(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t(b, \delta, i)} |e_0 h_0 + e_1 h_1| = |\mathcal{E}_t(b, \delta, i)| \cdot W_\nu(n, w, t), \end{aligned}$$

with

$$\begin{aligned} W_\nu(n, w, t) &= (n/2 - w + \nu) \cdot S(n-2, w, t-2) + \nu \cdot S(n-2, w-2, t-2) \\ &\quad + (w - 2\nu) \cdot (1 - S(n-2, w-1, t-2)). \end{aligned}$$

Proof. First remark that for any (b, δ) , each $(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t$ will appear in exactly $\mu_{\mathbf{e}_b}(\delta)$ of the sets $\mathcal{E}_t(b, \delta, i)$, $0 \leq i < r$, hence (i) and (ii) hold.

To $(h_0, h_1) \in \mathcal{H}_w$ we associate the matrix $\mathbf{H}$ consisting of two $r \times r$ circulant blocks, hence $n = 2r$, whose first columns are respectively h_0 and h_1 . To $(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t$ we associate the vector $\mathbf{e} \in \mathbb{F}_2^n$ obtained by concatenating $\mathbf{e}_0$ and $\mathbf{e}_1$. Indeed, we have $\mathbf{e}_0 h_0 + \mathbf{e}_1 h_1 = \mathbf{e}\mathbf{H}^\top$, and

$$\begin{aligned} \sum_{(\mathbf{e}_0, \mathbf{e}_1) \in \mathcal{E}_t(b, \delta, i)} |e_0 h_0 + e_1 h_1| &= |\mathcal{E}_t(b, \delta, i)| \cdot \mathbb{E} \left[|e_0 h_0 + e_1 h_1| \mid (\mathbf{e}_0, \mathbf{e}_1) \leftarrow^{\$} \mathcal{E}_t(b, \delta, i) \right] \\ &= |\mathcal{E}_t(b, \delta, i)| \cdot \mathbb{E} \left[\left| \mathbf{e}\mathbf{H}^\top \right| \mid \mathbf{e} \leftarrow^{\$} \mathcal{W}_{n, t}, e_j = e_{j'} = 1 \right], \end{aligned}$$

with $j = br + i$ and $j' = br + (i + \delta \bmod r)$. For given (b, δ) , we apply formula (17) of Lemma 3 to the matrix $\mathbf{H}$ and to the indices (j, j') . This formula makes use of the following coefficients

$$\nu_\ell = |\{i, 0 \leq i < r \mid |(h_{b,i}, h_{b,i+\delta})| = \ell\}|, 0 \leq \ell < 2.$$

By definition of the spectrum multiplicity, we have $\nu_2 = \nu = \mu_{h_b}(\delta)$. We must have $\nu_1 = w - 2\nu$ (the total weight of two columns is $2d = w$) and thus $\nu_0 = r - \nu_1 - \nu_2 = n/2 - w + \nu$. Finally, this yields

$$\sum_{(e_0, e_1) \in \mathcal{E}_t(b, \delta, i)} |e_0 h_0 + e_1 h_1| = |\mathcal{E}_t(b, \delta, i)| \cdot W_\nu(n, w, t).$$

□

We recall the proposition.

Proposition 3. *For all $(h_0, h_1) \in \mathcal{H}_w$ and all δ , $1 \leq \delta \leq r/2$, we have*

$$\frac{\sum_{(e_0, e_1) \in \mathcal{E}_t} \mu_{e_0}(\delta) |e_0 h_0 + e_1 h_1|}{\sum_{(e_0, e_1) \in \mathcal{E}_t} \mu_{e_0}(\delta)} = W_{\mu_{h_0}(\delta)}(n, w, t), \quad (6)$$

where for all integer $\nu \geq 0$

$$W_\nu(n, w, t) = (n/2 - w + \nu) \cdot S(n-2, w, t-2) + \nu \cdot S(n-2, w-2, t-2) \\ + (w - 2\nu) \cdot (1 - S(n-2, w-1, t-2)), \quad (7)$$

with

$$S(n, w, t) = \sum_{i \text{ odd}} \frac{\binom{w}{i} \binom{n-w}{t-i}}{\binom{n}{t}}.$$

Proof (of Proposition 3). We apply Lemma 4. First adding (iii) for all r , then use (i) and (ii) to get the identity (6) of Proposition 3.

B Proof of Proposition 4

The sum of two words of weight w and w' has a weight $w + w' - 2ww'/r$ when everything is uniformly distributed. The following Lemmas, which we admit, generalizes this for any distributions assuming some uniformity properties.

Lemma 5. *Let $\mathcal{D}$ denote a distribution over $\mathbb{F}_2^r$ such that the probability of $h \in \mathbb{F}_2^r$ only depends on its Hamming weight. For all $g \in \mathcal{R}$*

$$\mathbb{E} \left[|g + h| \mid h \xleftarrow{\mathcal{D}} \mathbb{F}_2^r \right] = \omega + (1 - 2\omega/r) \cdot |g|$$

where $\omega = \mathbb{E} \left[|h| \mid h \xleftarrow{\mathcal{D}} \mathbb{F}_2^r \right]$.

Lemma 6. *Let $\mathcal{D}_1$ and $\mathcal{D}_2$ denote distributions over $\mathbb{F}_2^r$ such that the probability of $h \in \mathbb{F}_2^r$ only depends on its Hamming weight. If h_1 and h_2 are drawn according*

to $\mathcal{D}_1$ and $\mathcal{D}_2$, the probability of $h_1 + h_2$ only depends on its Hamming weight. Moreover we have

$$\mathbb{E} \left[|h_1 + h_2| \mid h_1 \xleftarrow{\mathcal{D}_1} \mathbb{F}_2^r, h_2 \xleftarrow{\mathcal{D}_2} \mathbb{F}_2^r \right] = \omega_1 + (1 - 2\omega_1/r) \cdot \omega_2$$

where $\omega_b = \mathbb{E} \left[|h| \mid h \xleftarrow{\mathcal{D}_b} \mathbb{F}_2^r \right]$, $b \in \{1, 2\}$.

We recall the proposition.

Proposition 4. For all $(x, y) \in \mathcal{R}_\omega^2$ and all δ , $1 \leq \delta \leq r/2$, we have

$$\frac{\sum_{(r_1, r_2, e) \in \mathcal{M}} \mu_{r_2}(\delta) |r_1 y + r_2 x + e|}{\sum_{(r_1, r_2, e) \in \mathcal{M}} \mu_{r_2}(\delta)} = W_{\mu_x(\delta)}^{\text{HQC}}(r, \omega, \omega_r, \omega_e), \quad (12)$$

where for all integer $\nu \geq 0$

$$W_\nu^{\text{HQC}}(r, \omega, \omega_r, \omega_e) = \omega' + (1 - 2\omega'/r) \cdot \omega_1 \quad (13)$$

with

$$\begin{aligned} \omega_1 = (r - 2\omega + \nu) \cdot S(r - 2, \omega, \omega_r - 2) + \nu \cdot S(r - 2, \omega, \omega_r - 2) \\ + 2(\omega - \nu) \cdot (1 - S(r - 2, \omega, \omega_r - 2)) \end{aligned} \quad (14)$$

and

$$\omega' = \omega_e + (r - 2\omega_e) \cdot S(r, \omega, \omega_r). \quad (15)$$

Proof (of Proposition 4). Let $(x, y) \in \mathcal{R}_\omega^2$ and $(r_1, r_2, e) \in \mathcal{R}_{\omega_r}^2 \times \mathcal{R}_{\omega_e}$. We define $e' = r_1 y + r_2 x + e$. If r_1 is uniformly distributed, the distribution of $r_1 y$ verifies the uniformity condition of Lemma 6 and has an average weight $r \cdot S(r, \omega, \omega_r)$. From Lemma 6, the distribution of $r_1 y + e$ only depends on its Hamming weight and has an average weight $\omega' = \omega_e + (r - 2\omega_e) \cdot S(r, \omega, \omega_r)$.

Applying Lemma 5 with $r_1 y + e$ and a fixed value of $r_2 x$, we get

$$\mathbb{E} \left[|r_1 y + r_2 x + e| \mid r_1 \xleftarrow{\$} \mathcal{R}_{\omega_r}, e \xleftarrow{\$} \mathcal{R}_{\omega_e} \right] = \omega' + (1 - 2\omega'/r) \cdot |x r_2|. \quad (20)$$

Finally, a reasoning similar to the proof of Lemma 4 shows that for all δ

$$\frac{\sum_{r_2 \in \mathcal{R}_{\omega_r}} \mu_{r_2}(\delta) |r_2 x|}{\sum_{r_2 \in \mathcal{R}_{\omega_r}} \mu_{r_2}(\delta)} = \omega_1, \quad (21)$$

$$\begin{aligned} \omega_1 = (r - 2\omega + \nu) \cdot S(r - 2, \omega, \omega_r - 2) + \nu \cdot S(r - 2, \omega, \omega_r - 2) \\ + 2(\omega - \nu) \cdot (1 - S(r - 2, \omega, \omega_r - 2)). \end{aligned}$$

From (20) and (21), we easily derive

$$\frac{\sum_{(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}) \in \mathcal{M}} \mu_{\mathbf{r}_2}(\delta) |\mathbf{r}_2 \mathbf{x} + \mathbf{r}_1 \mathbf{y} + \mathbf{e}|}{\sum_{(\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}) \in \mathcal{M}} \mu_{\mathbf{r}_2}(\delta)} = \omega' + (1 - 2\omega'/r) \cdot \omega_1.$$

□